Pizza Sales Analysis using SQL

Presented by:

Saranya Koduru

Problem Statement

Analyze pizza sales data to identify revenue-driving products, customer ordering patterns, category performance, and opportunities for improving business decisions using SQL-based data analysis.

Dataset Overview

Below are the table names and their respective columns

orders : Stores overall customer order information including order date and time

Columns : order_id, order_date, order_time

order_details : Stores item-level details for each order

Columns : order_details_id, order_id, pizza_id, quantity

pizzas : Contains pizza size and pricing information for each pizza

Columns : pizza_id, pizza_type_id, size, price

pizza_types : Contains each pizza product details

Columns : pizza_type_id, name, category, ingredients

SQL Concepts Used

- Joins
- Aggregate Functions
- Group By & Having
- Window Functions
- Subqueries
- Common Table Expressions (CTEs)
- Revenue Calculations
- Ranking & Partitioning
- Date & Time Analysis
- Data Aggregation

Business Questions Analyzed

Basic Analysis

- Total orders and total revenue
- Highest-priced pizza
- Most common pizza size
- Top ordered pizzas by quantity

Intermediate Analysis

- Category-wise quantity distribution
- Order trends by hour and date
- Average pizzas ordered per day
- Top pizzas by revenue

Advanced Analysis

- Revenue contribution by pizza type
- Cumulative revenue analysis
- Top revenue-generating pizzas within each category

Additional Exploratory Analysis

- Total pizza variants in each category
- Revenue, order contribution by pizza variant
- Pizza-wise analysis by category, size, price, quantity, and revenue ranking

Key Business Metrics

Total Orders	Total Revenue
21,350	8,17,860.05

Common Pizza Size Ordered	Avg Pizzas / day
L	138

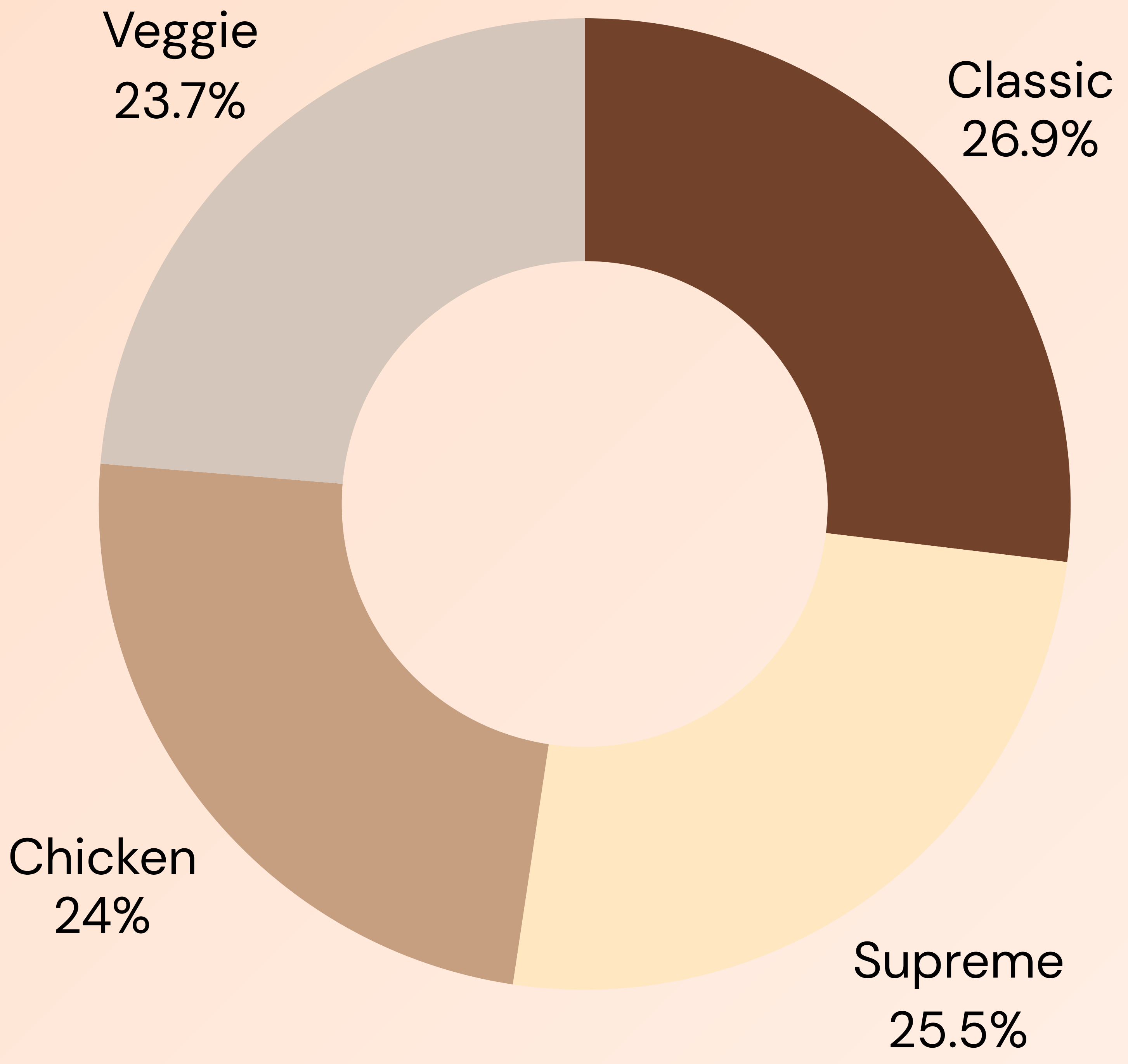
Highest - Priced Pizza

Category	Name	Price
Classic	The Greek Pizza	35.95

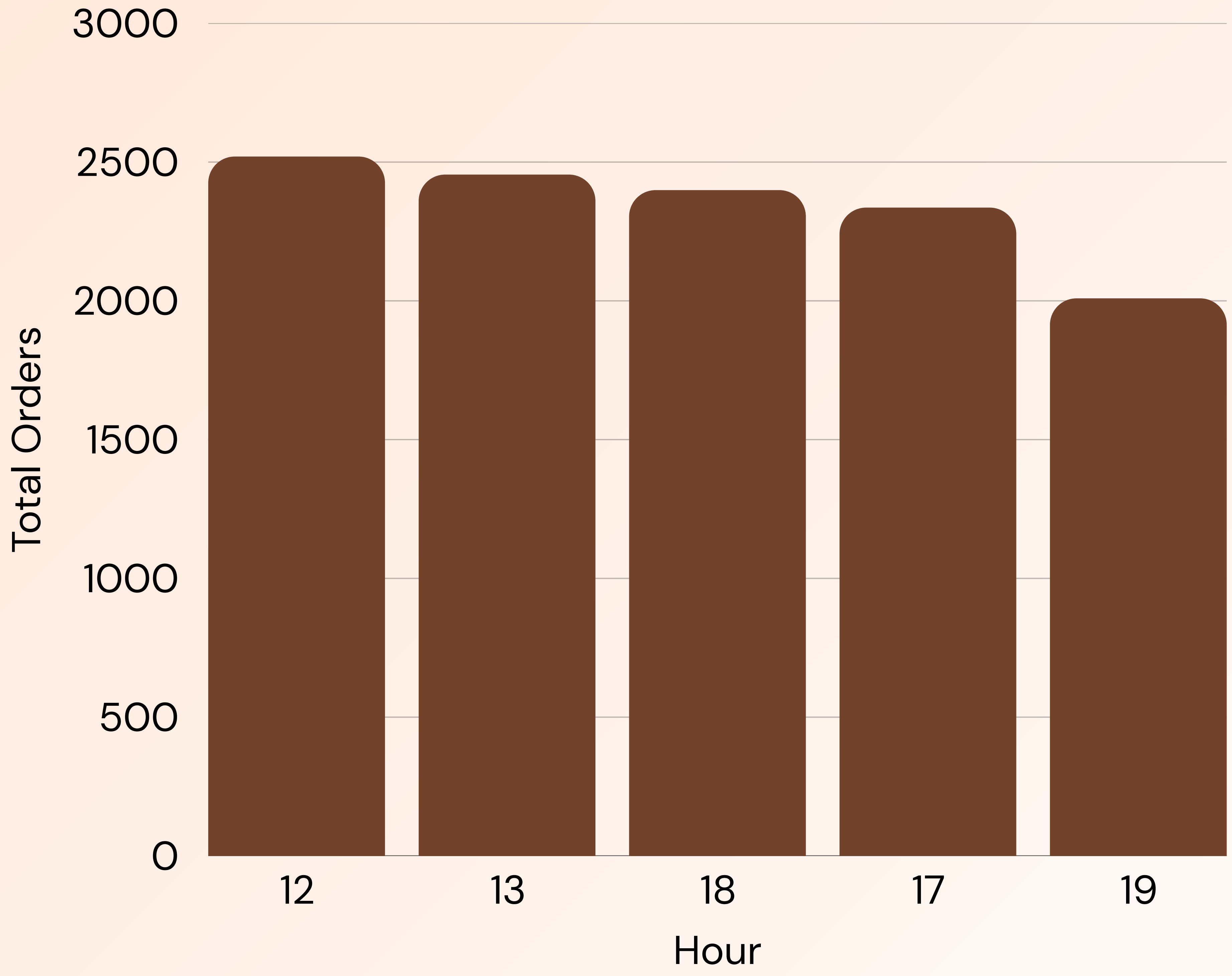
Top 5 most ordered pizza types & quantities

Pizza_type_id	Tot_quantity
classic_dlx	2453
bbq_ckn	2432
hawaiian	2422
pepperoni	2418
thai_ckn	2371

Category - Revenue Contribution



Top Most Ordered Hours



Pizzas with no sales recorded.

Category	pizza_type_id	Size	Price	Revenue	Total Quantity	Revenue Rank
Classic	big_meat	L	20.5	NULL	NULL	25
Classic	big_meat	M	16	NULL	NULL	25
Veggie	five_cheese	S	12.5	NULL	NULL	25
Veggie	five_cheese	M	15.5	NULL	NULL	25
Veggie	four_cheese	S	11.75	NULL	NULL	25
Classic	big_meat	L	20.5	NULL	NULL	25
Classic	big_meat	M	16	NULL	NULL	25
Veggie	five_cheese	S	12.5	NULL	NULL	25

Pizza Sales Performance Overview by Category

Category	pizza_type_id	Size	Price	Revenue	Total Quantity	Revenue Rank
Chicken	thai_ckn	L	20.75	29257.5	1410	1
Chicken	southw_ckn	L	20.75	21082	1016	2
Classic	big_meat	S	12	22968	1914	1
Classic	classic_dlx	M	16	18896	1181	2
Supreme	spicy_ital	L	20.75	23011.75	1109	1
Supreme	ital_supr	M	16.5	15526.5	941	2
Veggie	five_cheese	L	18.5	26066.5	1409	1
Veggie	four_cheese	L	17.95	23622.2	1316	2

Insights based on Analysis

- Classic category is stable with balanced sales and strong mass-market demand at lower prices.
- Supreme category performs consistently well despite higher prices, showing strong premium demand.
- Chicken and Veggie depend on a few top pizzas, while Classic and Supreme are more evenly spread.
- Large-size pizzas contribute most to revenue across categories, showing high customer preference.
- Some pizza-size combinations have zero sales and may need review or removal.

Recommendations

- Selling more Classic pizzas by giving simple combo offers and discounts can be a good option because people already like them.
- Improving Supreme pizzas by adding small extras and checking prices since demand is strong.
- Depending only on a few Chicken and Veggie pizzas wouldn't be a correct option. So promoting other pizzas with offers is a good strategy
- Encouraging customers to buy large size pizzas because they give most of the revenue.
- Removing or changing pizzas with no sales can be recommended to make the menu better.

Conclusion

Special Thanks ⭐

This is a guided project by Ayushi Jain from WsCubeTech. Along with the guided content, additional self-framed analysis questions were developed to explore deeper insights, leading to more detailed analysis and meaningful business recommendations.

[Guided Project Link]

Project Repository: [Github]